MATH221 Quiz 2, Fall 2025

KU Email: _____

Name (Korean or English): Solutions

Student ID # _____

Date	Monday, November 24th
Time Period	3:40 - 4:10pm
Duration of Quiz	30 minutes
Number of Test Pages	4 pages (2 double-sided sheets) in total
Total Possible Marks	24 points
Additional Materials Allowed	None. NO CALCULATORS.

Instructions

1. Answer the questions in the spaces provided. If you require additional space to answer a question, please use the extra page(s) and refer to this page in your solutions.
2. Your grade will be influenced by how clearly you express your ideas, and how well you organize your solutions.
3. **This quiz will be scanned and marked electronically.** Be sure to write legibly and **make sure that your writing isn't too faint / light.**
4. DO NOT WRITE NOTES/FORMULAS ON THIS COVER PAGE. If you do, you may get ZERO.
5. Writing any language other than **ENGLISH** on a solution will result **ZERO** on that problem. Writing other languages on a scratch work is okay, but you need to indicate which work is a scratch work and which is not.
6. Convention: column vectors can be rewritten as a tuple (not a row vector), i.e.,

$$\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} = (a_1, a_2, \ldots, a_n) \neq \begin{bmatrix} a_1 & a_2 & \cdots & a_n \end{bmatrix}.$$

1. (a) (6 points) Find inverse of the following invertible matrix

$$B = \begin{bmatrix} 3 & 2 \\ 5 & 3 \end{bmatrix}.$$

Sol: B^{-1} is soln of $[B \mid \mathbb{1}_2]$ via G-E.

$$[B \mid \mathbb{1}_2] = \left[\begin{array}{cc|cc} 3 & 2 & 1 & 0 \\ 5 & 3 & 0 & 1 \end{array}\right] \xrightarrow{R_1 \mapsto \frac{1}{3}R_1} \left[\begin{array}{cc|cc} 1 & \frac{2}{3} & \frac{1}{3} & 0 \\ 5 & 3 & 0 & 1 \end{array}\right] \xrightarrow{R_2 \mapsto R_2 - 5R_1} \left[\begin{array}{cc|cc} 1 & \frac{2}{3} & \frac{1}{3} & 0 \\ 0 & -\frac{1}{3} & -\frac{5}{3} & 1 \end{array}\right]$$

$$\xrightarrow{R_2 \mapsto -3R_2} \left[\begin{array}{cc|cc} 1 & \frac{2}{3} & \frac{1}{3} & 0 \\ 0 & 1 & 5 & -3 \end{array}\right] \xrightarrow{R_1 \mapsto R_1 - \frac{2}{3}R_2} \left[\begin{array}{cc|cc} 1 & 0 & -3 & 2 \\ 0 & 1 & 5 & -3 \end{array}\right]$$

$$\Rightarrow B^{-1} = \boxed{\begin{bmatrix} -3 & 2 \\ 5 & -3 \end{bmatrix}}$$

(b) (6 points) Suppose that $\beta = (e_1, e_2)$ is the standard basis of $\mathbb{R}^2$ and $\gamma = ((3,5),(2,3))$ is another basis of $\mathbb{R}^2$. Consider the following matrix

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}.$$

Use part (a) from above to compute $[L_A]_\gamma$.

Sol: $[\mathbb{1}_{\mathbb{R}^2}]_\gamma^\beta = B = \begin{bmatrix} 3 & 2 \\ 5 & 3 \end{bmatrix}$, so $[\mathbb{1}_{\mathbb{R}^2}]_\beta^\gamma = B^{-1} = \begin{bmatrix} -3 & 2 \\ 5 & -3 \end{bmatrix}$ from (a).

$$[L_A]_\gamma = [\mathbb{1}_{\mathbb{R}^2}]_\beta^\gamma [L_A]_\beta [\mathbb{1}_{\mathbb{R}^2}]_\gamma^\beta = B^{-1} A B = \begin{bmatrix} -3 & 2 \\ 5 & -3 \end{bmatrix}\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 3 & 2 \\ 5 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} -3\cdot 2 + 2\cdot 1 & -3\cdot 1 + 2\cdot 1 \\ 5\cdot 2 - 3\cdot 1 & 5\cdot 1 - 3\cdot 1 \end{bmatrix}\begin{bmatrix} 3 & 2 \\ 5 & 3 \end{bmatrix} = \begin{bmatrix} -4 & -1 \\ 7 & 2 \end{bmatrix}\begin{bmatrix} 3 & 2 \\ 5 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} -4\cdot 3 - 1\cdot 5 & -4\cdot 2 - 1\cdot 3 \\ 7\cdot 3 + 2\cdot 5 & 7\cdot 2 + 2\cdot 3 \end{bmatrix} = \boxed{\begin{bmatrix} -17 & -11 \\ 31 & 20 \end{bmatrix}}$$

2. (12 points) Use Gaussian-Elimination to check whether the following collection of vectors is linearly independent in $\mathbb{R}^4$ or not:

$$\beta := \{\underbrace{(1,1,2,2)}_{v_1}, \underbrace{(1,2,5,6)}_{v_2}, \underbrace{(1,3,6,8)}_{v_3}, \underbrace{(1,2,4,5)}_{v_4}\}.$$

Sol:

β lin indep $\Leftrightarrow$ only soln of $\sum_{i=1}^{4} a_i v_i = 0$ is $a_1 = \cdots = a_4 = 0 \overset{\text{R-N}}{\Leftrightarrow} \underbrace{\operatorname{rank}\begin{bmatrix} | & & | \\ v_1 & \cdots & v_4 \\ | & & | \end{bmatrix} = 4}_{\text{use G-E to compute this}}$

$$\begin{bmatrix} | & & | \\ v_1 & \cdots & v_4 \\ | & & | \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 2 \\ 2 & 5 & 6 & 4 \\ 2 & 6 & 8 & 5 \end{bmatrix} \overset{\substack{R_2 \mapsto R_2 - R_1 \\ R_3 \mapsto R_3 - 2R_1 \\ R_4 \mapsto R_4 - 2R_1}}{\rightsquigarrow} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 3 & 4 & 2 \\ 0 & 4 & 6 & 3 \end{bmatrix} \overset{\substack{R_3 \mapsto R_3 - 3R_2 \\ R_4 \mapsto R_4 - 4R_2}}{\rightsquigarrow} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & -2 & -1 \\ 0 & 0 & -2 & -1 \end{bmatrix}$$

$$\overset{R_3 \mapsto -\frac{1}{2}R_3}{\rightsquigarrow} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & \frac{1}{2} \\ 0 & 0 & -2 & -1 \end{bmatrix} \overset{R_4 \mapsto R_4 + 2R_3}{\rightsquigarrow} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

has rank ≤ 3 b/c 4th row is 0

$\Rightarrow \operatorname{rank}\left(\begin{bmatrix} | & & | \\ v_1 & \cdots & v_4 \\ | & & | \end{bmatrix}\right) < 4 \Rightarrow \beta = \{v_1, \ldots, v_4\}$ **lin dep**.

USE THIS PAGE IF ADDITIONAL SPACE IS REQUIRED
(DO NOT TEAR IT OFF)
Clearly state the question number being answered and refer the marker to this page.